Task 1: Install RHEL with LVM

During installation:

- Choose Custom under Installation Destination.
- Select LVM as the storage configuration.
- Complete the installation.
- Log in to the system.

Task 2: Verify Root Partition Uses LVM

Display block devices:

- lsblk

Display volume groups:

- vgs

Display logical volumes:

- lvs

Display physical volumes:

- pvs

Task 3: Extend the Root Logical Volume

Steps

1. Verify the new disk
    - lsblk
2. Create a Physical Volume
    - sudo pvcreate /dev/sdb
3. Find the Volume Group Name
    - vgs
4. Extend the Volume Group
    - sudo vgextend rhel /dev/sdb
5. Check disk usage before extending
    - df -h /
6. Extend the Root Logical Volume by 1 GB
    - sudo lvextend -L +1G /dev/rhel/root
7. Resize the Filesystem
    - For ext4:
        - sudo resize2fs /dev/rhel/root

    - For XFS (default in RHEL):
        - sudo xfs_growfs /
8. Verify
    - df -h /

Task 4: Create a New Logical Volume

Create a 500 MB logical volume:

- sudo lvcreate -L 500M -n musicdata rhel

Format the logical volume.
For ext4:

- sudo mkfs.ext4 /dev/rhel/musicdata

Create the mount point:

- sudo mkdir /mnt/music

Mount it:

- sudo mount /dev/rhel/musicdata /mnt/music

Configure Automatic Mounting
Find the UUID:

- sudo blkid /dev/rhel/musicdata

Edit the fstab file:

- sudo vi /etc/fstab

Add this line:

- UUID=1234-5678-ABCD   /mnt/music   ext4   defaults   0 0

Save and exit.

Test the configuration:

- sudo umount /mnt/music
- sudo mount -a